Algoritmo de Simon

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Estrategia General de los Algoritmos cuánticos

- Iniciar los qubits en un estado clásico.
- Poner los qubits en superposición
- Realizar las operaciones en los qubits.
- Realizar una observación

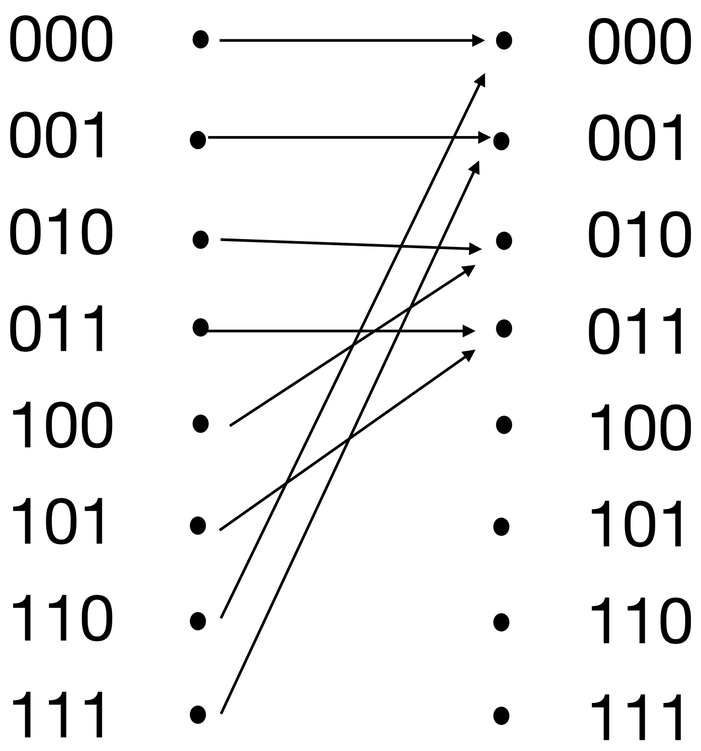
Problema que deseamos resolver

- Imagine que le dan una caja negra que implementa una función de $\{0,1\}^n$ hacia $\{0,1\}^n$.
- Es decir, $f : \{0,1\}^n \to \{0,1\}^n$
- Le dicen adicionalmente que existe una cadena binaria secreta $c = c_0 c_1 c_2 \ldots c_{n-1}$, dónde cada c_j es un 0 o un 1. Es decir $c_j \in \{0,1\}$.
- Tal que para todas las cadenas $x, y \in \{0,1\}^n$, tenemos que:
  - $f(x) = f(y)$ si solo si $x = y \oplus c$
  - Dónde $\oplus$ es un operación or-exclusivo bit a bit
- c es llamado el **periodo** de f.
- **El objetivo del algoritmo es determinar el periodo, es decir determinar c.**

Ejemplo de función periódica

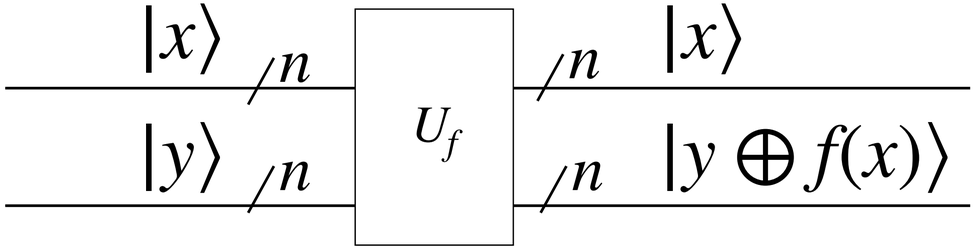
Ejercicio: Determine f si n = 3 y c = 110.

Una de las funciones que satisface la condición:

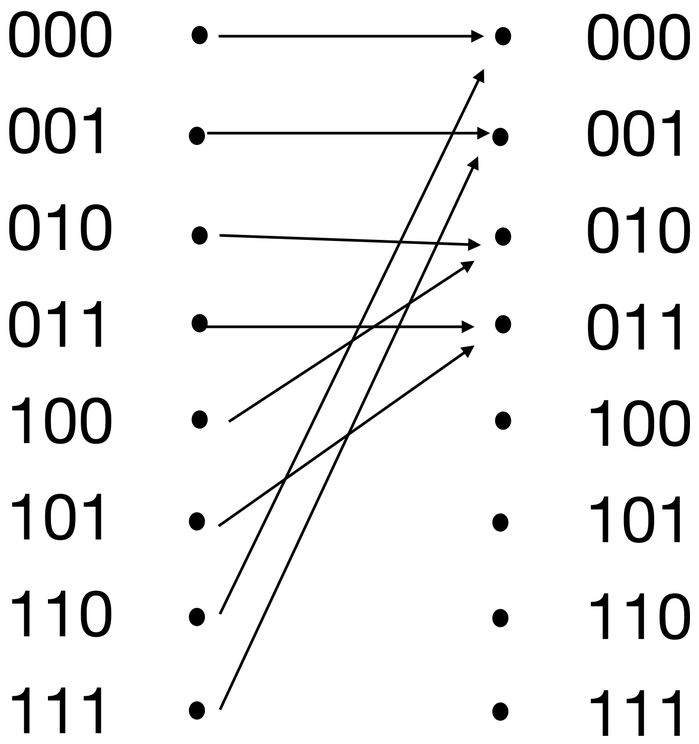


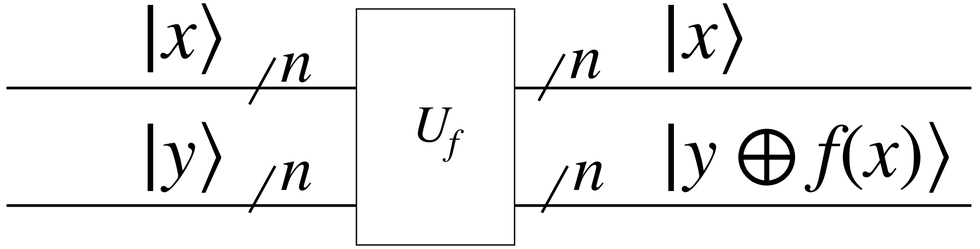
- $f : \{0,1\}^3 \rightarrow \{0,1\}^3$
- $f(x) = f(y)$ si solo si $x = y \oplus c$
  - $000 \oplus 110 = 110$; esto implica que: $f(000) = f(110)$
  - $001 \oplus 110 = 111$; esto implica que: $f(001) = f(111)$
  - $010 \oplus 110 = 100$; esto implica que: $f(010) = f(100)$
  - $011 \oplus 110 = 101$; esto implica que: $f(011) = f(101)$
  - $100 \oplus 110 = 010$; esto implica que: $f(100) = f(010)$
  - $101 \oplus 110 = 011$; esto implica que: $f(101) = f(011)$
  - $110 \oplus 110 = 000$; esto implica que: $f(110) = f(000)$
  - $111 \oplus 110 = 001$; esto implica que: $f(111) = f(001)$

¿Podemos representar el problema con compuertas y matrices?



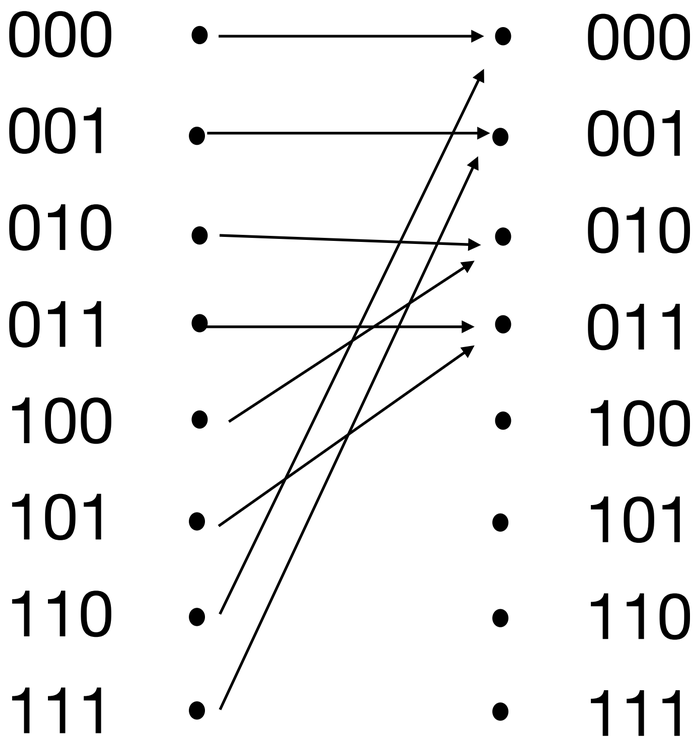
¿Podemos representar el problema con compuertas y matrices?

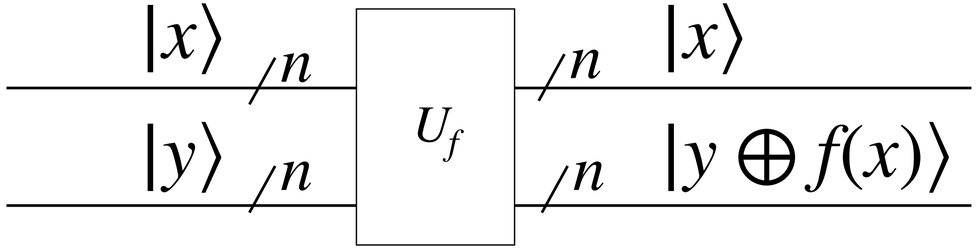




	000, 000	000,001	000,010	000,011	…	010,100	010,101	010,110	010,111	…
000, 000					…					…
000,001					…					…
000,010					…					…
000,011					…					…
⋮					…					…
010,100					…					…
010,101					…					…
010,110					…					…
010,111					…					…
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

¿Podemos representar el problema con compuertas y matrices?





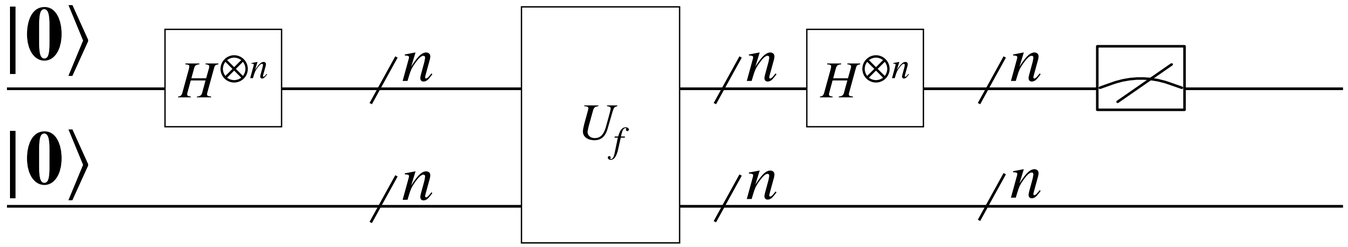
	000, 000	000,001	000,010	000,011	…	010,100	010,101	010,110	010,111	…
000, 000	1				…					…
000,001		1			…					…
000,010			1		…					…
000,011				1	…					…
⋮					…					…
010,100					…			1		…
010,101					…				1	…
010,110					…	1				…
010,111					…		1			…
⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮	⋮

Vamos a resolverlo en computador clásico

- Si $c = 0^n$ entonces la función es uno a uno (i.e., cada elemento del dominio envía una flecha a un elemento diferente de la imagen)
- Si $c \neq 0^n$ entonces la función es dos a uno (i.e., tiene periodo cada dos elementos del dominio envían flecha a un elemento de la imagen).
- Podemos buscar dos valores de x tal que:
  - $f(x_1) = f(x_2)$
- Una vez tengamos dos valores entonces podemos calcular c así:
  - $x_1 \oplus x_2 = x_2 \oplus c \oplus x_2 = c$
- Si la función es dos a uno al evaluar la mitad más uno ya debería haber una coincidencia. Si evaluamos más de la mitad más uno sin coincidencia podemos concluir que la función es uno a uno y $c = 0^n$
- Peor caso: $\frac{2^n}{2} + 1 = 2^{n-1} + 1$

¿Podemos hacerlo mejor con un sistema cuántico?

El algoritmo



$$|\psi_3\rangle = (H^{\otimes n} \otimes I) U_f (H^{\otimes n} \otimes I) |\psi_0\rangle$$

El algoritmo funciona?

Recordatorio de producto tensor de H

$$H^{\otimes n}[i,j] = \frac{1}{\sqrt{2}^{n}}(-1)^{\langle i,j\rangle}$$

Sea $x = x_0x_1x_2x_3\ldots x_{n-1}$, $y = y_0y_1y_2y_3\ldots y_{n-1}$ dos cadenas de la misma longitud y

$$\langle , \rangle : \{0,1\}^n \times \{0,1\}^n \to \{0,1\} : \langle x,y\rangle = x_0 \wedge y_0 \oplus x_1 \wedge y_1 \oplus \ldots \oplus x_{n-1} \wedge y_{n-1}$$

Esta función determina la paridad de los 1 que coinciden, es decir si coinciden un número par de veces o impar de veces. Cuando el número de coincidencias es par, el exponente es 0, y cuando el número de coincidencias es impar, el exponente es 1.

Si $y \in \{0,1\}^n$ $|y\rangle$ es un vector columna con un solo 1 en alguna posición. Al multiplicar lo que hago es extraer la columna y de la matriz.

$$H^{\otimes n}|y\rangle = H^{\otimes n}[\,-\,,y] = \frac{1}{\sqrt{2}^{n}} \sum_{x\in\{0,1\}^n} (-1)^{\langle x,y\rangle}|x\rangle$$

Recordatorio de propiedades de $\otimes$

$$(A \otimes B)^\dagger = A^\dagger \otimes B^\dagger$$

$$(A \star A') \otimes (B \star B') = (A \otimes B) \star (A' \otimes B')$$

$$(A \otimes B) \star (V \otimes V') = (A \star V) \otimes (B \star V')$$

Producto tensor respeta adición en $\mathbb{V}$ y en $\mathbb{V}'$

$$(V_i + V_j) \otimes V'_k = V_i \otimes V'_k +'' V_j \otimes V'_k$$

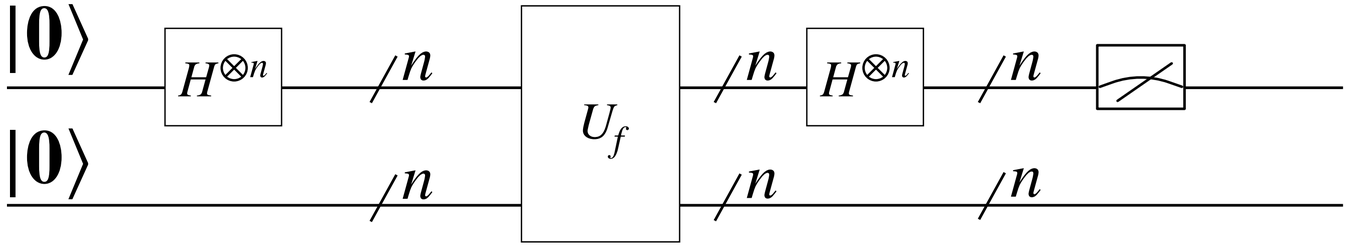
$$V_i \otimes (V'_j +' V'_k) = V_i \otimes V'_j +'' V_i \otimes V'_k$$

Producto tensor respeta la multiplicación escalar en $\mathbb{V}$ y en $\mathbb{V}'$

$$c \cdot'' (V_j \otimes V'_k) = (c \cdot V_j) \otimes V'_k = V_j \otimes (c \cdot' V'_k)$$

El algoritmo de Simon

Paso 1: $|\psi_0\rangle$



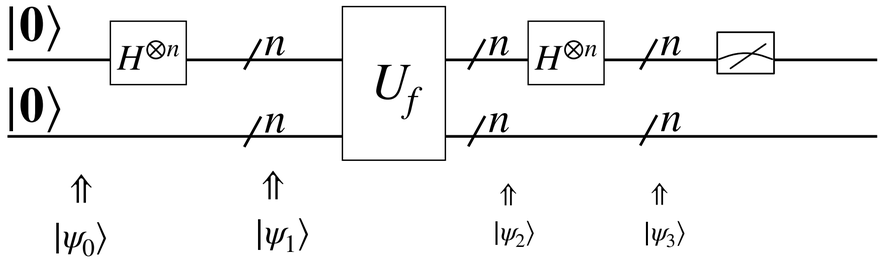
$|\psi_0\rangle$ $|\psi_1\rangle$ $|\psi_2\rangle$ $|\psi_3\rangle$

$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$

El algoritmo de Simon

Paso 2: $|\psi_1\rangle$

$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$



$$|\psi_1\rangle = H^{\otimes n} * |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = [\frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x\rangle] \otimes |\mathbf{0}\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle$$

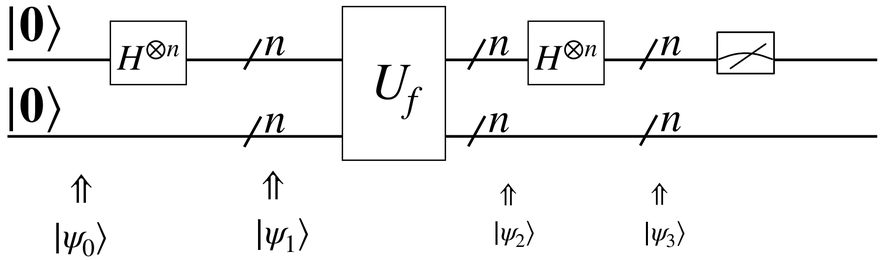
$$\frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle = \frac{1}{\sqrt{2}^n}(|000...0,000...0\rangle + |0000...001,000...0\rangle + |0000...010,000...0\rangle + |0000...011,000...0\rangle + |0000...100,000...0\rangle + \ldots + |1111...110,000...0\rangle + |1111...111,000...0\rangle)$$

El algoritmo de Simon

Paso 3: $|\psi_2\rangle$

$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$

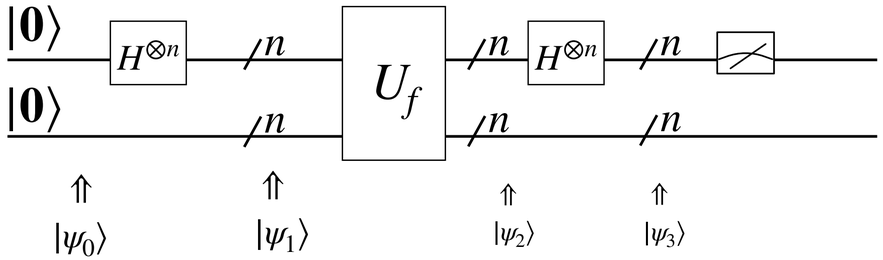
$$|\psi_1\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle$$



$$|\psi_2\rangle = [\frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,f(x)\rangle]$$

El algoritmo de Simon

Paso 4: $|\psi_3\rangle$



$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,f(x)\rangle$$

$$|\psi_3\rangle = (H^{\otimes n} \otimes I) * \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x\rangle \otimes |f(x)\rangle$$

$$|\psi_3\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} [(H^{\otimes n} \otimes I) * |x\rangle \otimes |f(x)\rangle] = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} [H^{\otimes n} * |x\rangle \otimes I * |f(x)\rangle]$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} \sum_{z\in\{0,1\}^n} (-1)^{\langle z,x\rangle} |z\rangle \otimes |f(x)\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} \sum_{z\in\{0,1\}^n} (-1)^{\langle z,x\rangle} |z,f(x)\rangle$$

El algoritmo de Simon

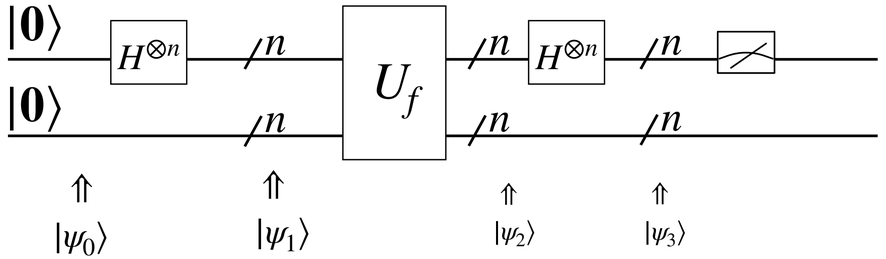
Paso 4: $|\psi_3\rangle$

$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,f(x)\rangle$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} \sum_{z\in\{0,1\}^n} (-1)^{\langle z,x\rangle} |z,f(x)\rangle$$



Por las características de la función usted sabe que para cada z y para cada x, el ket $|z,f(x)\rangle$ es igual al ket $|z,f(x\oplus c)\rangle$. El coeficiente de ese ket es:

$$\frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\otimes c\rangle}}{2}$$

El coeficiente lo podemos reescribir como:

$$\frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\oplus c\rangle}}{2} = \frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\rangle\oplus\langle z,c\rangle}}{2}$$

$$= \frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\rangle}(-1)^{\langle z,c\rangle}}{2}$$

Si $\langle z,c\rangle = 1$ los términos se cancelan y el coeficiente es 0.

Si $\langle z,c\rangle = 0$ los términos se suman y el coeficiente es +/-1.

El algoritmo de Simon

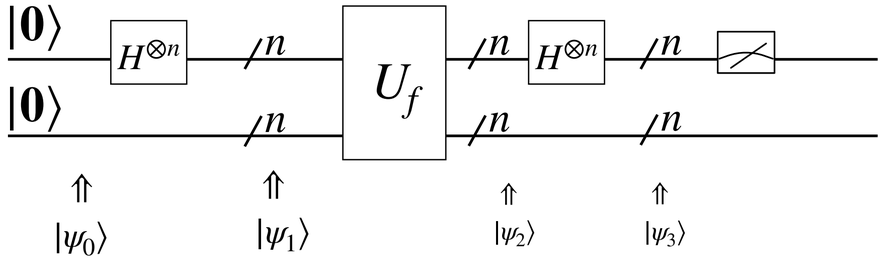
Paso 4: $|\psi_3\rangle$

$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |\mathbf{0},\mathbf{0}\rangle = |00000...000,00000...000\rangle = [1,0,0,0,0,...,0]^T$$

$$|\psi_1\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,\mathbf{0}\rangle$$

$$|\psi_2\rangle = \frac{1}{\sqrt{2}^n} \sum_{x\in\{0,1\}^n} |x,f(x)\rangle$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} \sum_{z\in\{0,1\}^n} (-1)^{\langle z,x\rangle} |z,f(x)\rangle$$



El coeficiente lo podemos reescribir como:

$$\frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\oplus c\rangle}}{2} = \frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\rangle\oplus\langle z,c\rangle}}{2}$$

$$= \frac{(-1)^{\langle z,x\rangle} + (-1)^{\langle z,x\rangle}(-1)^{\langle z,c\rangle}}{2}$$

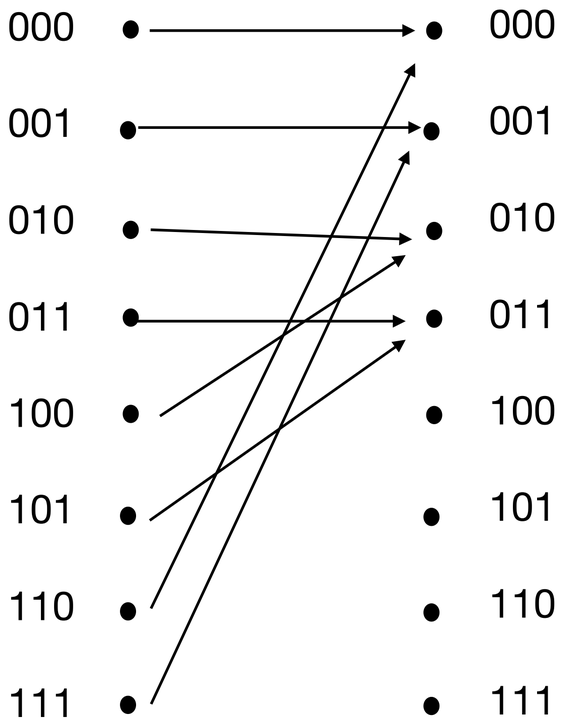
Si $\langle z,c\rangle = 1$ los términos se cancelan y el coeficiente es 0.

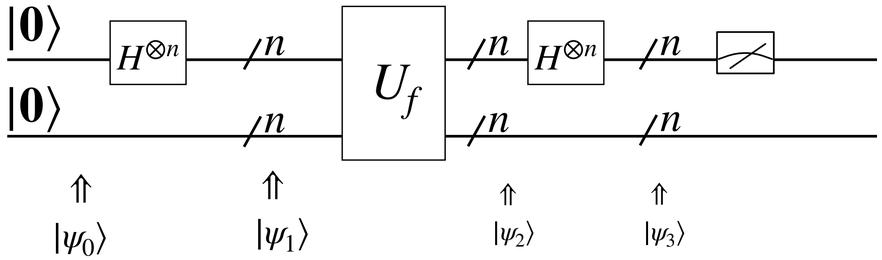
Si $\langle z,c\rangle = 0$ los términos se suman y el coeficiente es 1.

Por lo tanto al medir los qubits de arriba solo vamos a encontrar cadenas binarias tal que $\langle z,c\rangle = 0$.

Una vez tiene las cadenas z que cumplen esto, usted puede despejar c.

Ejemplo





$$|\psi_0\rangle = |\mathbf{0}\rangle \otimes |\mathbf{0}\rangle = |000,000\rangle$$

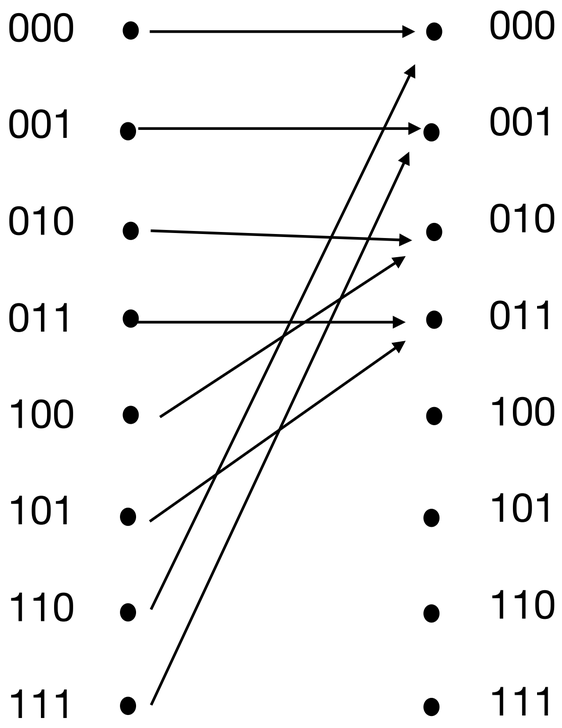
$$|\psi_1\rangle = \frac{1}{\sqrt{8}} \sum_{x\in\{0,1\}^n} |x, \mathbf{0}\rangle$$

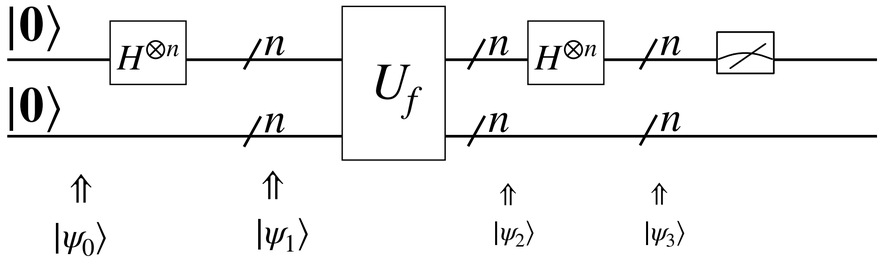
$$|\psi_1\rangle = \frac{1}{\sqrt{8}}(|000,000\rangle + |001,000\rangle + |010,000\rangle + |011,000\rangle +$$
$$|100,000\rangle + |101,000\rangle + |110,000\rangle + |111,000\rangle)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{8}}(|000,f(000)\rangle + |001,f(001)\rangle + |010,f(010)\rangle + |011,f(011)\rangle +$$
$$|100,f(100)\rangle + |101,f(101)\rangle + |110,f(110)\rangle + |111,f(111)\rangle)$$

$$|\psi_2\rangle = \frac{1}{\sqrt{8}}(|000,000\rangle + |001,001\rangle + |010,010\rangle + |011,011\rangle +$$
$$|100,010\rangle + |101,011\rangle + |110,000\rangle + |111,001\rangle)$$

Ejemplo

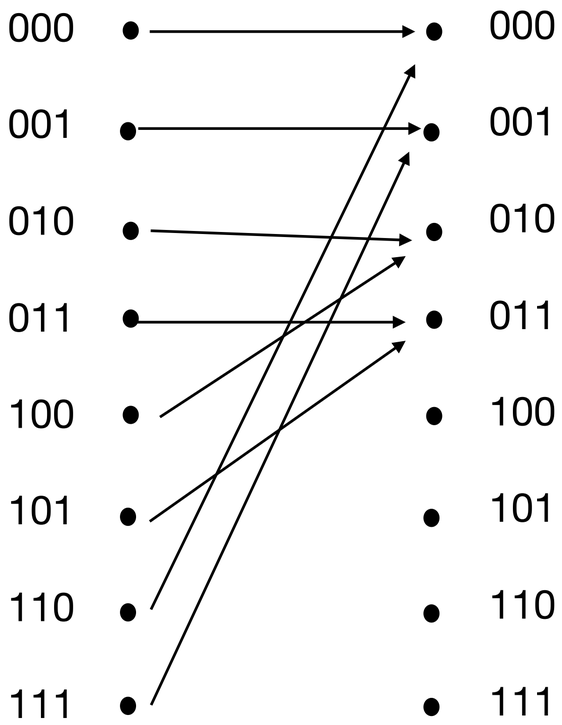


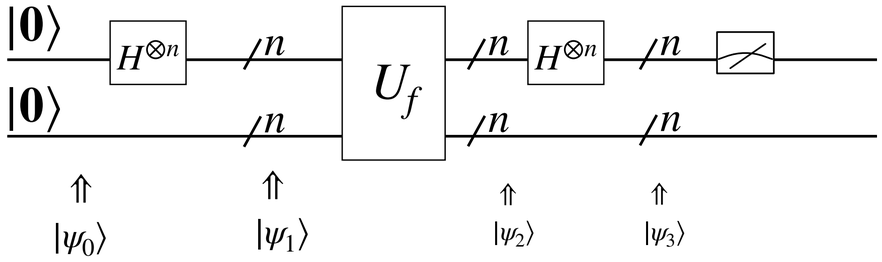


$$|\psi_2\rangle = \frac{1}{\sqrt{8}}(|000,f(000)\rangle + |001,f(001)\rangle + |010,f(010)\rangle + |011,f(011)\rangle +$$
$$|100,f(100)\rangle + |101,f(101)\rangle + |110,f(110)\rangle + |111,f(111)\rangle)$$

$$|\psi_3\rangle = \frac{1}{2^n} \sum_{x\in\{0,1\}^n} \sum_{z\in\{0,1\}^n} (-1)^{\langle z,x\rangle} |z, f(x)\rangle$$

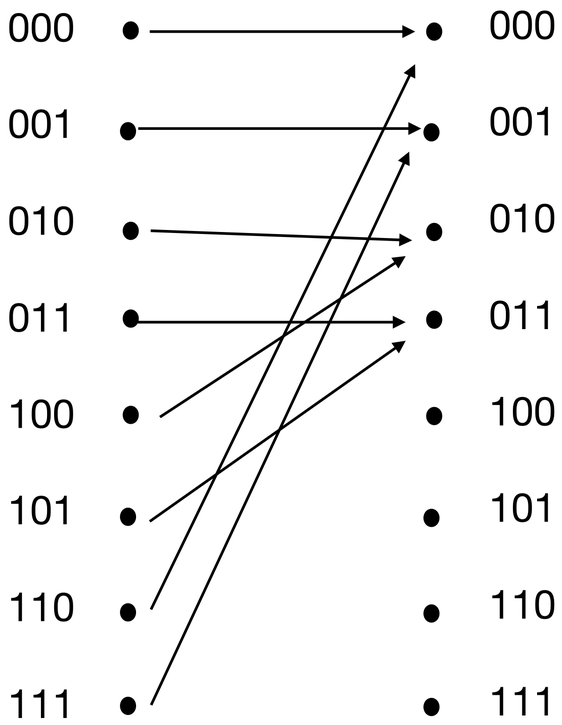
Ejemplo

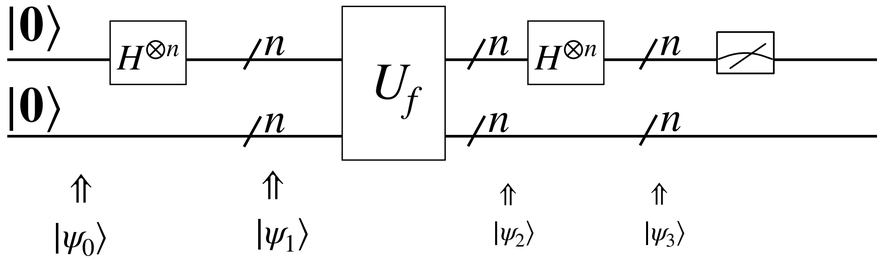




$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{8}(& \\
& ((1)|000,f(000)\rangle + (1)|001,f(000)\rangle + (1)|010,f(000)\rangle + (1)|011,f(000)\rangle + \\
& (1)|100,f(000)\rangle + (1)|101,f(000)\rangle + (1)|110,f(000)\rangle + (1)|111,f(000)\rangle) + \\
& ((1)|000,f(001)\rangle + (-1)|001,f(001)\rangle + (1)|010,f(001)\rangle + (-1)|011,f(001)\rangle + \\
& (1)|100,f(001)\rangle + (-1)|101,f(001)\rangle + (1)|110,f(001)\rangle + (-1)|111,f(001)\rangle) + \\
& ((1)|000,f(010)\rangle + (1)|001,f(010)\rangle + (-1)|010,f(010)\rangle + (-1)|011,f(010)\rangle + \\
& (1)|100,f(010)\rangle + (1)|101,f(010)\rangle + (-1)|110,f(010)\rangle + (-1)|111,f(010)\rangle) + \\
& ((1)|000,f(011)\rangle + (-1)|001,f(011)\rangle + (-1)|010,f(011)\rangle + (1)|011,f(011)\rangle + \\
& (1)|100,f(011)\rangle + (-1)|101,f(011)\rangle + (-1)|110,f(011)\rangle + (1)|111,f(011)\rangle) + \\
& ((1)|000,f(100)\rangle + (1)|001,f(100)\rangle + (1)|010,f(100)\rangle + (1)|011,f(100)\rangle + \\
& (-1)|100,f(100)\rangle + (-1)|101,f(100)\rangle + (-1)|110,f(100)\rangle + (-1)|111,f(100)\rangle) + \\
& ((1)|000,f(101)\rangle + (-1)|001,f(101)\rangle + (1)|010,f(101)\rangle + (-1)|011,f(101)\rangle + \\
& (-1)|100,f(101)\rangle + (1)|101,f(101)\rangle + (-1)|110,f(101)\rangle + (1)|111,f(101)\rangle) + \\
& ((1)|000,f(110)\rangle + (1)|001,f(110)\rangle + (-1)|010,f(110)\rangle + (-1)|011,f(110)\rangle + \\
& (-1)|100,f(110)\rangle + (-1)|101,f(110)\rangle + (1)|110,f(110)\rangle + (1)|111,f(110)\rangle) + \\
& ((1)|000,f(111)\rangle + (-1)|001,f(111)\rangle + (-1)|010,f(111)\rangle + (1)|011,f(111)\rangle + \\
& (-1)|100,f(111)\rangle + (1)|101,f(111)\rangle + (1)|110,f(111)\rangle + (-1)|111,f(111)\rangle))
\end{aligned}
$$

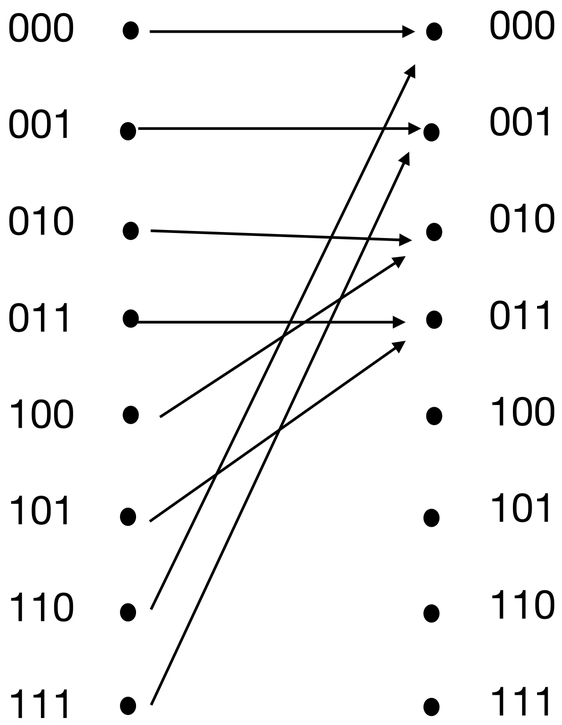
Ejemplo

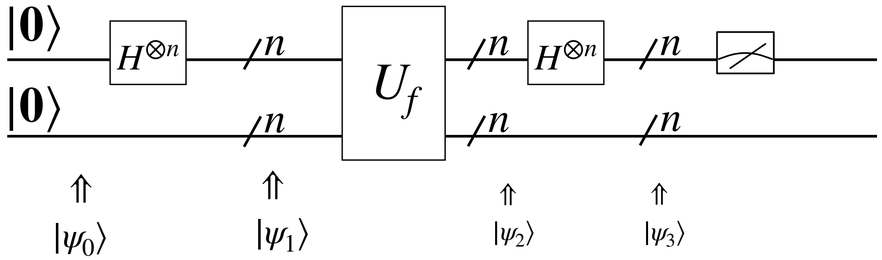




$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{8}(& \\
& ((1)|000,f(000)\rangle + (1)|001,f(000)\rangle + (1)|010,f(000)\rangle + (1)|011,f(000)\rangle + \\
& (1)|100,f(000)\rangle + (1)|101,f(000)\rangle + (1)|110,f(000)\rangle + (1)|111,f(000)\rangle) + \\
& ((1)|000,f(001)\rangle + (-1)|001,f(001)\rangle + (1)|010,f(001)\rangle + (-1)|011,f(001)\rangle + \\
& (1)|100,f(001)\rangle + (-1)|101,f(001)\rangle + (1)|110,f(001)\rangle + (-1)|111,f(001)\rangle) + \\
& ((1)|000,f(010)\rangle + (1)|001,f(010)\rangle + (-1)|010,f(010)\rangle + (-1)|011,f(010)\rangle + \\
& (1)|100,f(010)\rangle + (1)|101,f(010)\rangle + (-1)|110,f(010)\rangle + (-1)|111,f(010)\rangle) + \\
& ((1)|000,f(011)\rangle + (-1)|001,f(011)\rangle + (-1)|010,f(011)\rangle + (1)|011,f(011)\rangle + \\
& (1)|100,f(011)\rangle + (-1)|101,f(011)\rangle + (-1)|110,f(011)\rangle + (1)|111,f(011)\rangle) + \\
& ((1)|000,f(100)\rangle + (1)|001,f(100)\rangle + (1)|010,f(100)\rangle + (1)|011,f(100)\rangle + \\
& (-1)|100,f(100)\rangle + (-1)|101,f(100)\rangle + (-1)|110,f(100)\rangle + (-1)|111,f(100)\rangle) + \\
& ((1)|000,f(101)\rangle + (-1)|001,f(101)\rangle + (1)|010,f(101)\rangle + (-1)|011,f(101)\rangle + \\
& (-1)|100,f(101)\rangle + (1)|101,f(101)\rangle + (-1)|110,f(101)\rangle + (1)|111,f(101)\rangle) + \\
& ((1)|000,f(110)\rangle + (1)|001,f(110)\rangle + (-1)|010,f(110)\rangle + (-1)|011,f(110)\rangle + \\
& (-1)|100,f(110)\rangle + (-1)|101,f(110)\rangle + (1)|110,f(110)\rangle + (1)|111,f(110)\rangle) + \\
& ((1)|000,f(111)\rangle + (-1)|001,f(111)\rangle + (-1)|010,f(111)\rangle + (1)|011,f(111)\rangle + \\
& (-1)|100,f(111)\rangle + (1)|101,f(111)\rangle + (1)|110,f(111)\rangle + (-1)|111,f(111)\rangle))
\end{aligned}
$$

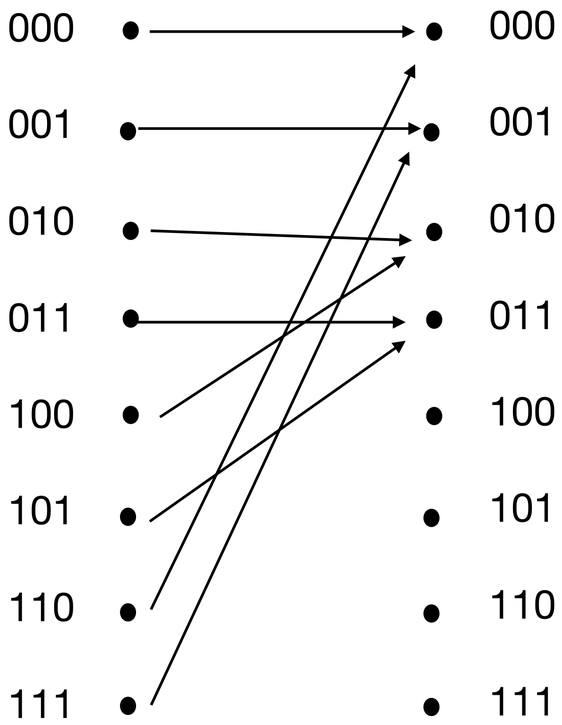
Ejemplo



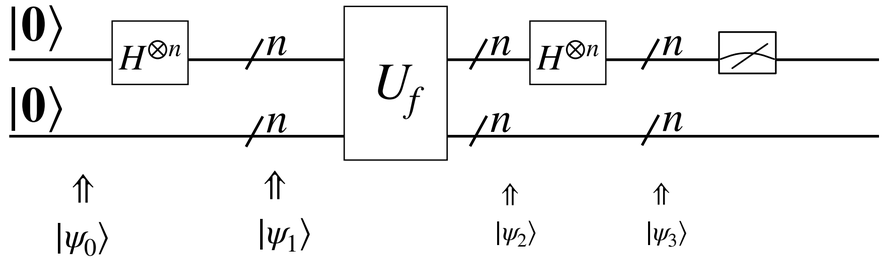


$$
\begin{aligned}
|\psi_3\rangle = \frac{1}{8}(\\
&(1)|000,f(000)\rangle + (1)|000,f(110)\rangle + \\
&(1)|110,f(000)\rangle + (1)|110,f(110)\rangle + \\
&(1)|000,f(001)\rangle + (1)|000,f(111)\rangle + \\
&(1)|110,f(001)\rangle + (1)|110,f(111)\rangle + \\
&(1)|000,f(010)\rangle + (1)|000,f(100)\rangle + \\
&(-1)|110,f(010)\rangle + (-1)|110,f(100)\rangle + \\
&(1)|000,f(011)\rangle + (1)|000,f(101)\rangle + \\
&(-1)|110,f(011)\rangle + (-1)|110,f(101)\rangle + \\
&(1)|001,f(010)\rangle + (1)|001,f(100)\rangle + \\
&(-1)|111,f(010)\rangle(-1)|111,f(100)\rangle + \\
&(-1)|001,f(011)\rangle + (-1)|001,f(101)\rangle + \\
&(1)|111,f(011)\rangle + (1)|111,f(101)\rangle + \\
&(1)|001,f(000)\rangle + (1)|001,f(110)\rangle + \\
&(1)|111,f(000)\rangle(1)|111,f(110)\rangle + \\
&(-1)|001,f(001)\rangle + (-1)|001,f(111)\rangle + \\
&(-1)|111,f(001)\rangle + (-1)|111,f(111)\rangle \\
)
\end{aligned}
$$

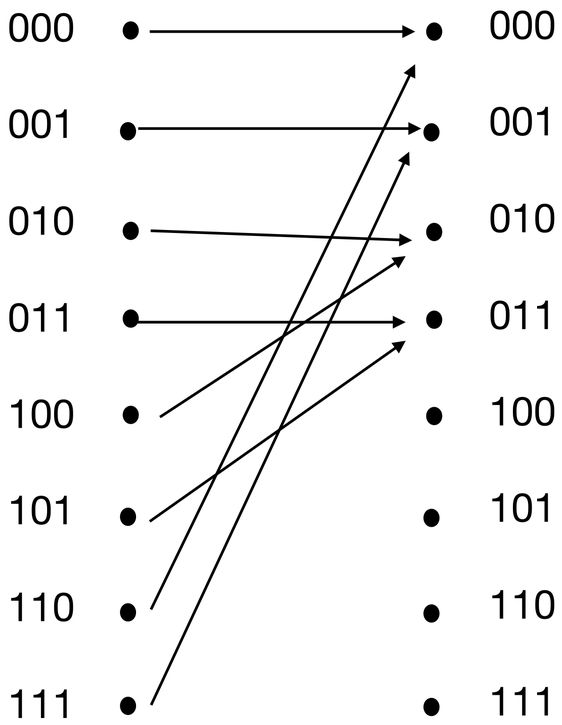
Ejemplo

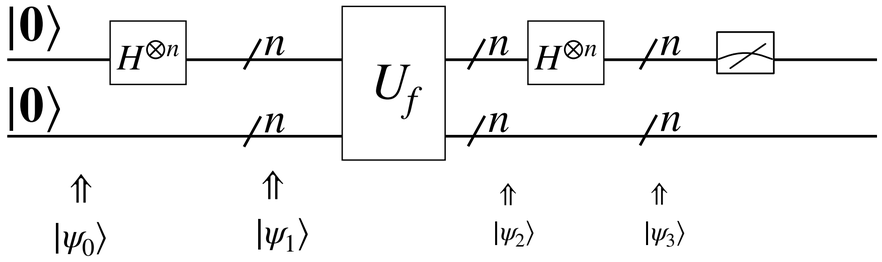


$$|\psi_3\rangle = \frac{1}{8}($$
$$(2)|000,000\rangle +$$
$$(2)|110,000\rangle +$$
$$(2)|000,001\rangle +$$
$$(2)|110,001\rangle +$$
$$(2)|000,010\rangle +$$
$$(-2)|110,010\rangle +$$
$$(2)|000,011\rangle +$$
$$(-2)|110,011\rangle +$$
$$(2)|001,010\rangle +$$
$$(-2)|111,010\rangle +$$
$$(-2)|001,011\rangle +$$
$$(2)|111,011\rangle +$$
$$(2)|001,000\rangle +$$
$$(2)|111,000\rangle +$$
$$(-2)|001,001\rangle +$$
$$(-2)|111,001\rangle +$$
$$)$$



Ejemplo

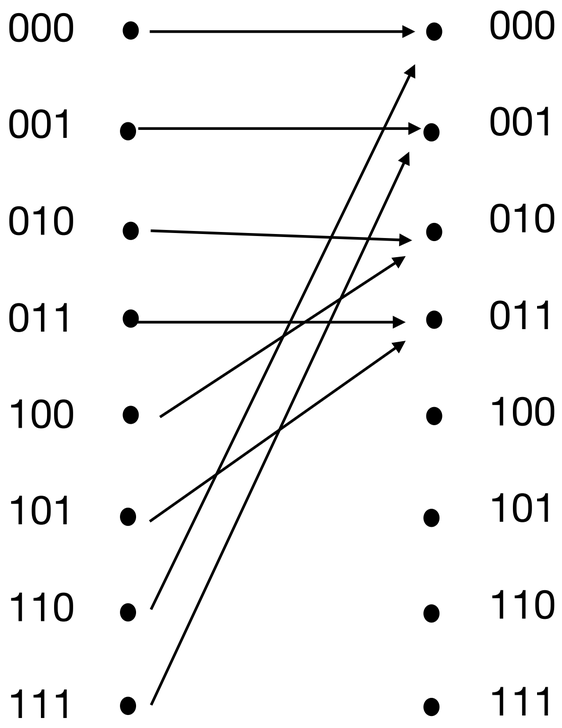


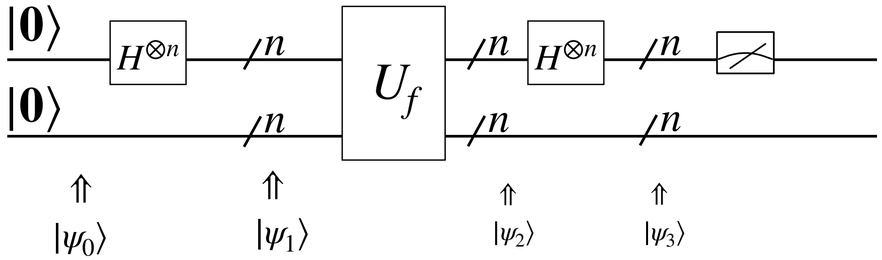


$$|\psi_3\rangle = \frac{1}{8}($$
$$(2)|000,000\rangle + (2)|000,001\rangle + (2)|000,010\rangle + (2)|000,011\rangle +$$
$$(2)|110,000\rangle + (2)|110,001\rangle + (-2)|110,010\rangle + (-2)|110,011\rangle +$$
$$(2)|001,010\rangle + (-2)|001,011\rangle + (2)|001,000\rangle + (-2)|001,001\rangle +$$
$$(-2)|111,010\rangle + (2)|111,011\rangle + (2)|111,000\rangle + (-2)|111,001\rangle$$
$$)$$

$$|\psi_3\rangle = \frac{1}{8}($$
$$(2 * |000\rangle \otimes (|000\rangle + |001\rangle + |010\rangle + |011\rangle)) +$$
$$(2 * |110\rangle \otimes (|000\rangle + |001\rangle - |010\rangle - |011\rangle)) +$$
$$(2 * |001\rangle \otimes (|010\rangle - |011\rangle + |000\rangle - |001\rangle)) +$$
$$(2 * |111\rangle \otimes (- |010\rangle + |011\rangle + |000\rangle - |001\rangle))$$
$$)$$

Ejemplo





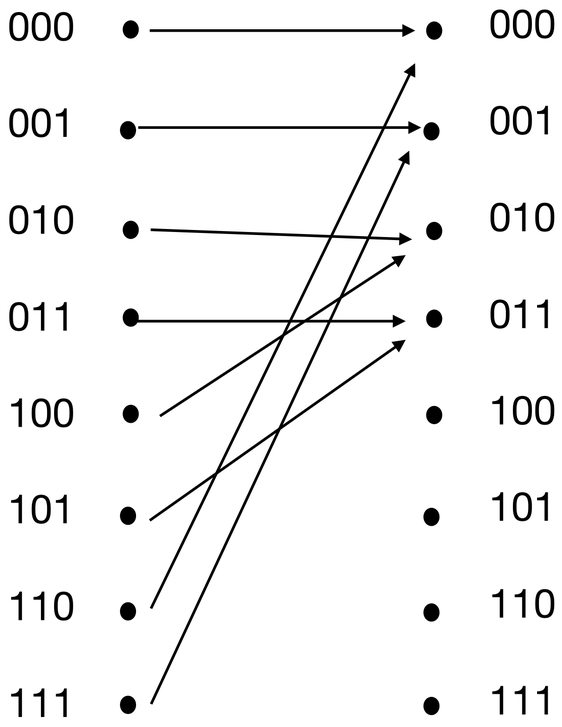
$$|\psi_3\rangle = \frac{1}{8}($$
$$(2 * |000\rangle \otimes (|000\rangle + |001\rangle + |010\rangle + |011\rangle)) +$$
$$(2 * |110\rangle \otimes (|000\rangle + |001\rangle - |010\rangle - |011\rangle)) +$$
$$(2 * |001\rangle \otimes (|010\rangle - |011\rangle + |000\rangle - |001\rangle)) +$$
$$(2 * |111\rangle \otimes (-|010\rangle + |011\rangle + |000\rangle - |001\rangle))$$
$$)$$

Las cadenas $|000\rangle, |001\rangle, |110\rangle, |111\rangle$ son las que cumplen con $\langle z, c\rangle = 0$.

Si suponemos que $c = c_0c_1c_2$ con cada c_j un valor de 0 o 1, podemos despejar c así:

Ejemplo





Las cadenas $|000\rangle, |001\rangle, |110\rangle, |111\rangle$ son las que cumplen con $\langle z, c\rangle = 0$.

Si suponemos que $c = c_0c_1c_2$ con cada c_j un valor de 0 o 1, podemos despejar c así:

$i)$ $\langle 000, c\rangle = 0$

$ii)$ $\langle 001, c\rangle = 0$

$iii)$ $\langle 110, c\rangle = 0$

$vi)$ $\langle 111, c\rangle = 0$

La ecuación ii nos dice que $c_2 = 0$, la ecuación iii nos dice que $c_0 \oplus c_1 = 0$ (es decir son ambos 0 o ambos 1), como sabemos que $c \neq 000$, entonces:

$$c = 110$$

¿Preguntas?